

# Lecture4

September 9, 2026

## 1 Lecture 4

```
[28]: # while loops
      # for loop has a certain number of iterations
      # while loop repeats until the condition is satisfied

      # find the largest integer who's cube is less than

      i = 0
      while (i**3 < 2026):
          i += 1

      print((i-1)**3)
```

1728

```
[20]: 13**3
```

```
[20]: 2197
```

```
[26]: 12**3
```

```
[26]: 1728
```

```
[2]: # break and continue
      # find if it is prime or not
      n = 2027

      # m is the divisor
      for m in range(2, n):
          if n%m ==0:
              print(f"{n} is not prime: {n} = {m} * {n//m}")
              break
          last = m

      if last == n - 1:
          print(n, 'is a prime')
```

2027 is a prime

```
[ ]: # Continue skips
```

```
i = 0  
while i < 10:  
    i += 1
```